BPUT
HACKATHON 2026

StackMind

# StackMind

***Smarter ESG. Stronger Tomorrow.***

## AI-Powered ESG Control Tower for BRSR Reporting

| | |
|---|---|
| **Team ID**<br>BH26PS08T003 | **Team Name**<br>STACKMIND |
| **Category**<br>Software | **Members**<br>5 |



5 Members | One Vision | Sustainable Future

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# Why MEIL Needs an ESG Control Tower?

## Key ESG Challenges

- **Data Fragmentation**
  ESG data is scattered across different sources, formats and systems.
- **Data Inconsistency**
  Data comes from multiple formats, sources and manual submissions.
- **Data Duplication**
  Repeated data entry can create duplicate and conflicting information.
- **Evidence Gap**
  Recorded ESG data is not always directly linked to supporting documents or proof.
- **Calculation Complexity**
  Different units, methodologies and emission factors make calculations complex.
- **Version Control**
  Emission factors, rules and calculation methods may change across reporting periods.
- **Manual Governance**
  Data validation, review and approvals depend heavily on manual effort.
- **Data Consolidation**
  ESG information from numerous operational locations must be combined.

## ESG Control Tower

A centralized, governed platform that connects data collection, validation, calculation, reporting and auditability.




Data Errors → Delayed Reporting → Difficult Validation → Weak Traceability → Higher Manual Effort → Complex BRSR Consolidation

Root Causes → Key ESG Challenges → Reporting Impact → Need for ESG Control Tower

The core challenge is not collecting ESG data—it is ensuring that every reported ESG number is accurate, verified, traceable, and reproducible from its original source

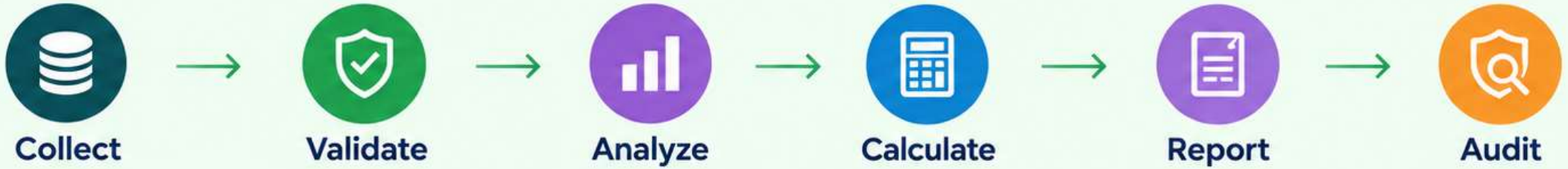
# Existing Solutions VS Our Proposed Solution

FROM MANUAL PROCESSES TO AN INTELLIGENT ESG PLATFORM

| Problems (Manual \| Fragmented \| Time Consuming) | | Our Proposed Solution (Automated \| Integrated \| Accurate) |
|---|---|---|
| Data from multiple sources (projects, bills, invoices, etc.) | Data Collection | Centralized digital data collection |
| Multiple Excel files & scattered data | Document Processing | OCR-based extraction from unstructured documents |
| Manual checking & reconciliation | Data Verification | Automated comparison & validation |
| Manual emission & ESG calculations | ESG Calculations | Automated calculation engine |
| Spreadsheet-based tracking | Carbon Emission | Automatic emission calculation |
| Difficult to identify manually | Resource & Wastage Tracking | Automated usage & wastage analysis |
| Manually prepared | Anomaly Detection | Automated anomaly detection |
| Scattered records | BRSR Report | System-generated BRSR report |
| Difficult to consolidate | Entity Management | Centralized entity-wise management |

# Workflow & Data Transparency

## End-to-End ESG Data Flow




## Detailed Workflow

| | | |
|---|---|---|
| 1 | Data Collection | Multi-source (APIs, files, manual) |
| 2 | Validation | Format, completeness, consistency |
| 3 | Analysis | Mapping, classification, anomaly detection |
| 4 | Calculation | Emissions, KPIs, ESG metrics |
| 5 | Reporting | BRSR + custom reports |
| 6 | Audit | Logs, version history, approvals |

## Data Transparency

- Real-time Tracking
- Audit Logs
- Version History
- Role-based Access
- Immutable Records

Transparent data. Trusted reporting.

MEIL
ESG PLATFORM

Dashboard | Sign Out

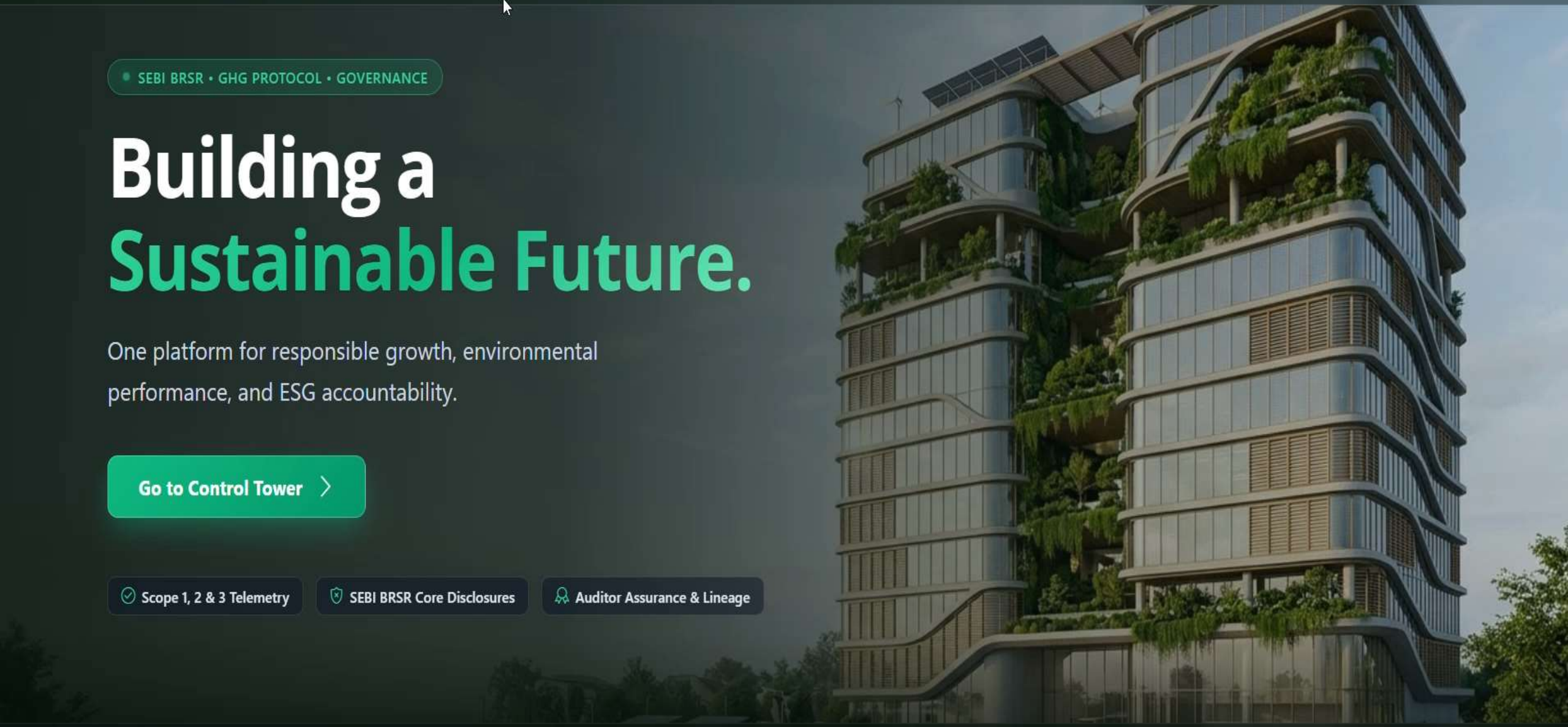
SEBI BRSR • GHG PROTOCOL • GOVERNANCE

# Building a Sustainable Future.

One platform for responsible growth, environmental performance, and ESG accountability.

Go to Control Tower

Scope 1, 2 & 3 Telemetry | SEBI BRSR Core Disclosures | Auditor Assurance & Lineage



© 2026 ESG Platform | Sustainability • Accountability • Transparency

# Role-Based ESG Workflow – Data to Validation

Different roles. Specific responsibilities. One unified ESG platform.

## 1 Site Manager
Data Capture

Collect and submit project-level information from the site.

**Workflow:**

Project Activity → Enter Data → Upload Documents → Submit

**Key Responsibilities:**

- Enters project activity data (fuel, water, type, suppliers, etc.)
- Uploads bills, invoices, and supporting documents
- Adds resource / employee information
- Reviews entered data for accuracy
- Submits project data for verification

**Submitted Data**

## 2 Subsidiary Manager
Review & Consolidation

Review data coming from multiple projects under the subsidiary and consolidate it for further processing.

**Workflow:**

Receive → Review → Compare → Approve/ Return

**Key Responsibilities:**

- Reviews project submissions
- Checks consistency with reported values
- Identifies missing or inconsistent information
- Sends corrections when required
- Approves candidate project data

**Validated & Consolidated Data**

## 3 ESG Manager
Calculation & Reporting

Convert validated data into ESG insights, calculate metrics and generate ESG reports.

**Workflow:**

Validated Data → Calculate → Analyze → Generate Report

**Key Responsibilities:**

- Calculates ESG metrics (energy, water, waste, etc.)
- Calculates carbon emissions
- Analyzes resource usage and savings
- Reviews anomalies
- Generates BRSR / ESG reports

**Finalized ESG Data & Reports**

## 4 Auditor
Verification & Audit Trail

Independently verify the reported ESG information and maintain a complete audit trail.

**Workflow:**

Review Submissions → Verify → Approve/ Flag → Audit Trail

**Key Responsibilities:**

- Reviews reported ESG metrics
- Checks supporting evidence (bills, docs, calculations)
- Verifies calculations and processes
- Flags discrepancies when found
- Maintains a complete audit history

**Verified & Audited Reports**

## Complete Role Flow

From data capture to audited ESG reports.

1. **Capture** — Site Manager — Project data + documents
2. **Review** — Subsidiary Manager — Validation & consolidation
3. **Calculate** — ESG Manager — Carbon metrics + BRSR
4. **Verify** — Auditor — Audit trail + approval

# Scope 1, 2 & 3 Emissions

Understanding the source. Measuring the impact.

*Cleaner Data → Greener Tomorrow*

## 1 Scope 1
### Direct Emissions

**Emissions from sources owned or controlled by the company.**



**Examples**

- Petrol and diesel burned in company vehicles.
- Fuel used in generators and project machinery.
- Natural gas burned in company facilities.

**Website role**

Records fuel consumption and calculates the resulting direct emissions.

## 2 Scope 2
### Indirect Energy Emissions

**Emissions generated when producing the electricity, steam, heating or cooling the company purchases and consumes.**



**Examples**

- Electricity purchased for offices.
- Electricity used at project sites.
- Purchased steam or cooling (if applicable).

**Website role**

Uses electricity bills and the right energy emission factor to estimate emissions.

## 3 Scope 3
### Other Value Chain Emissions

**All other indirect emissions occurring across the company's upstream and downstream value chain.**



**Examples**

- Emissions from purchased raw materials.
- Transportation by third-party suppliers.
- Business travel & employee commuting.
- Waste treatment by external providers.

**Website role**

Uses supplier, transport, travel and waste data to estimate relevant value-chain emissions.

## How the 3 Scopes Differ

| Scope | Emission Source | Simple Example |
|---|---|---|
| Scope 1 | Direct, company-controlled sources | Diesel burned in own generator |
| Scope 2 | Purchased energy generation | Electricity used in the office |
| Scope 3 | Upstream / downstream activities | Manufacturing purchased cement |

## How Our Website Calculates Emissions

Upload Bills & Activity Data → Validate Data & Select Emission Factors → Calculate (Scopes 1, 2 & 3) → Emission Dashboard & BRSR Reporting

**Carbon Emissions → Activity Data → Emission Factor**

**Example:** Litres of diesel consumed → Diesel emission factor = Estimated $CO_2e$

**Note:** Scope 3 can be the largest category for many businesses, but its size depends on the industry and activities. It's not a separate indicator, but a value-chain source.

## The Complete Emissions Flow

1. Upload Data
2. Validate & Select Factors
3. Calculate & Classify Scopes 1, 2 & 3
4. Dashboard & BRSR Reporting

Better Data → Lower Emissions → A Sustainable Future

## Carbon Emission Calculation

**Emissions → Activity Data × Emission Factor**

Diesel (litres) × Emission Factor (kg $CO_2e$/litre) = $CO_2e$ (kg or tonnes)

## Key Takeaways

- Lower pollution, better disclosure.
- Accurate data → reliable BRSR reporting.
- The right data makes a bigger impact.

*Measure → Manage → Make an Impact*

# Carbon Emissions Calculation • Process Flow • Technology Stack

From raw project data to verified ESG impact – powered by technology.

*Cleaner Projects Greener Tomorrow*

## 1. Carbon Emissions Calculation

**Emissions = Activity Data × Emission Factor**
(kg $CO_2e$)

### Illustrative Examples

* Factors are illustrative examples only

#### Diesel

| | |
|---|---|
| Fuel Consumed | 500 L |
| Emission Factor | 2.68 kg $CO_2e$/L |

**Emissions** **1,340 kg $CO_2e$** (1.34 t $CO_2e$)

#### Electricity

| | |
|---|---|
| Electricity Consumed | 2,000 kWh |
| Emission Factor | 0.70 kg $CO_2e$/kWh |

**Emissions** **1,400 kg $CO_2e$** (1.40 t $CO_2e$)

**Total Carbon Emissions** **2,740 kg $CO_2e$** (2.74 t $CO_2e$)

**Note:** Emission factors are illustrative examples and should be sourced from authoritative and up-to-date references for actual reporting.

## 2. Calculation Process & Flow

1. **Data Input**
   - Fuel bills
   - Electricity bills
   - Water usage
   - Waste records
   - Employee data
2. **OCR / Data Extraction**
   - Extract values
   - Read documents
   - Capture data
3. **Validation**
   - Compare with entered data
   - Flag discrepancies
   - Manual review
4. **Emissions Calculation**
   - Apply emission factors
   - Convert units
   - Calculate totals
5. **Scope-wise Aggregation**
   - Project level
   - Site level
   - Subsidiary level
   - Reporting period
6. **BRSR Dashboard & Reporting**
   - ESG indicators
   - Verified results
   - Supporting evidence
   - Audit trail

## 3. Technology Stack

**Django** (Backend Framework)
- Secure & scalable
- REST APIs
- Clean architecture

**Role-Based Access Control (RBAC)**
- Role management
- Secure authentication
- Access control

**ESG Calculation Engine**
- Emission factors
- Formula engine
- Unit conversion

**OCR-based Document Extraction**
- Extract data from bills
- Parse structured info
- Validate & match

**BRSR Reporting & Dashboard**
- ESG dashboard
- BRSR report generation
- Export (PDF/Excel)

**Confirm before presenting:** Specific **frontend**, **database**, **OCR provider** and **cloud vendors** are **not defined** in the current project scope and should be verified with the development team.

**Reliable data → Accurate calculations → Better sustainability reporting**

# FUTURE VISION

Making ESG reporting smarter, more secure and scalable.



## 01 Offline Data Entry & Automatic Sync

Allow users to enter project details, resource consumption and ESG data even without an internet connection. Save records locally and automatically upload pending data when the user reconnects to the internet.

**Benefit:** Uninterrupted data collection at remote project sites.



## 02 Advanced Data Security

Strengthen platform security through encrypted data storage and transmission, secure authentication, role-based access permissions and audit logs. Protect uploaded bills, employee information and ESG reports from unauthorized access.

**Benefit:** Better privacy, controlled access and reliable data tracking.




## 03 AI-Powered ESG Intelligence

Enhance the platform to identify unusual resource consumption, detect mismatches between bills and entered data, and provide useful insights into emissions, wastage and resource usage.

**Benefit:** Faster anomaly detection and more informed decisions.



## 04 Multi-Company Scalability

Expand the platform to support multiple companies, subsidiaries and project locations through centralized monitoring and consolidated ESG/BRSR reporting.

**Benefit:** A scalable platform for growing organizations.

**FROM MANUAL REPORTING TO INTELLIGENT, SECURE & SCALABLE ESG MANAGEMENT.**

# THANK YOU!

Making ESG Reporting Smarter, Simpler and More Transparent.

## STACKMIND

ESG & BRSR Reporting Platform




Presented by:
**Stack Mind Team**